Cycle Test 2: worked solutions

15 Oct 2025 · 20 marks · 60 min · CEPC11

Part A (1 mark each)

Q1

1 mark

Match the following.

	Property		Matches
A	Rapid slump loss	I	High fine content
B	High cohesiveness but low flowability	II	Pumpable concrete
C	Zero slump	III	High C_3A or hot weather
D	High cohesiveness, good flowability	IV	Dry mix concrete

A–III, B–I, C–IV, D–II.

- Rapid slump loss: fast early hydration (high C_3A) or high temperature.
- Many fines make a mix cohesive but stiff (high water demand, low flow).
- Zero slump: very dry mix (e.g. for pavers, roller-compacted concrete).
- Pumpable concrete must be cohesive (no segregation in the pipe) and flowable.

Q2

1 mark

Weak concrete layer on the surface due to segregation is known as _____.

Laitance: a weak layer of fine cement particles and water brought to the surface by bleeding/segregation.

Q3

1 mark

> Corrosion can be reduced if the pH of concrete is less than 9. (True/False)

False. Steel stays **passive** (protected by a stable oxide film) in the **high pH (about 12.5–13.5)** of concrete. When pH falls below about **11** (e.g. by carbonation), the film breaks down (**depassivation**) and corrosion starts. A pH below 9 makes corrosion **more** likely, not less.

Q4

1 mark

> Mention the application where internal sulphate attack is critical or a concern.

Internal sulfate attack, i.e. **delayed ettringite formation (DEF)**, happens when concrete reaches **more than about 70 °C at early age**, which prevents normal ettringite formation. Critical in:

- **steam-cured / heat-cured precast** products (e.g. **railway sleepers**, precast girders, pipes);
- **mass concrete** (thick rafts, dam sections, large piers) where heat of hydration raises the core temperature;
- **hot-weather concreting** of high-cement mixes.

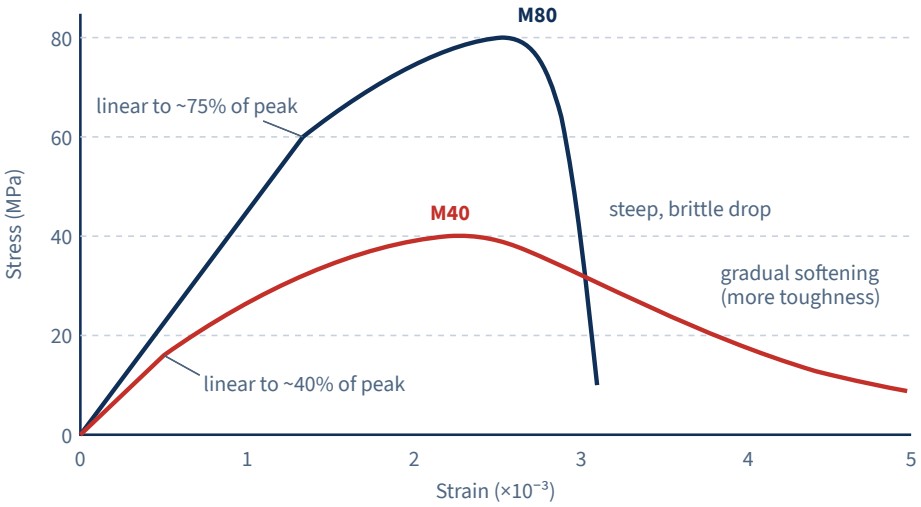
Part B (3 marks each)

Q5

3 marks

> Consider two types of concrete (say M40 and M80) tested under uniaxial compression.
>
> a. Draw the stress–strain curves indicating different zones of response.
>
> b. Compare and explain the difference in their key properties (in terms of pre-peak, peak and post-peak responses).

a)



Zones for each curve: **linear elastic** (bond cracks stable) → **non-linear pre-peak** (bond and matrix cracks grow) → **peak** → **post-peak softening** (cracks localize).

b)

	M40 (normal strength)	M80 (high strength)
Pre-peak	Linear only up to about **40%** of peak; more ITZ microcracking; lower E (about 32 GPa)	**Stiffer** (higher E, about 45 GPa); **longer linear portion** (about 70–80% of peak) because the denser paste and ITZ crack less
Peak	About 40 MPa at strain ≈ 0.002	About 80 MPa at slightly **higher strain** (≈ 0.0025–0.003)
Post-peak	**Gradual descending branch**; large area under the curve → more **ductile, tougher**	**Steep, sudden drop**; **brittle**, smaller strain at failure; low toughness

Reason: in HSC the paste and ITZ are so strong that cracks pass **through** the aggregate, so failure is sudden; in NSC cracks go round the aggregate along the weak ITZ, absorbing more energy.

Q6

3 marks

> a. Define and differentiate the shrinkage and creep mechanism.
>
> b. Brief the critical aspects of different types of shrinkage.

a) Both are **time-dependent strains** that involve **movement of water** in the cement paste.

	Shrinkage	Creep
Definition	Volume reduction **without load**, as water is lost (or consumed)	Gradual **increase of strain under a sustained load**
Cause of water movement	**Loss of water** to the surroundings or to hydration	Water **forced to move by stress**
Mechanism	Capillary tension in menisci, loss of adsorbed and interlayer water, C-S-H compaction	Water in C-S-H micropores diffuses from high-stress to low-stress zones; C-S-H particles slip and densify (viscoelastic)
Direction	**Volumetric** (same in all directions)	In the **direction of the stress** (with Poisson effect laterally)
Restrained form	Tensile stress → cracking	**Relaxation** (stress falls under constant strain)
Recoverable?	Partly (on re-wetting)	Partly (creep recovery)

b) Critical aspects of each type of shrinkage:

- **Plastic shrinkage**: water evaporates from the surface **before setting**. Critical in hot, windy weather and slabs with large surface area; causes surface cracks. Keep evaporation below **1 kg/m²/h**, cure early, use fibres or SRA.
- **Autogenous shrinkage**: chemical shrinkage (hydrates occupy 8–12% less volume than cement + water) plus **self-desiccation**, with no water lost to the outside. Critical at **low w/c (HSC)**: up to 700×10^{-6} at w/c 0.17.
- **Thermal contraction**: cooling after the heat of hydration peak. Critical in **mass concrete**.

- **Drying shrinkage**: loss of water from hardened concrete to dry air; largest long-term component, partly irreversible. Increases with paste content and w/c; reduced by aggregates.
- **Carbonation shrinkage**: CH dissolves and $CaCO_3$ deposits; maximum at **intermediate RH (about 50%)**.

Part C (10 marks)

Q7

10 marks

> As a concrete expert, design concrete for a **bridge deck member in a coastal region**. Assumptions can be made for any other relevant data; justify them.
>
> Grade M50; cement PPC conforming to IS 1489 (Part 1); GGBS conforming to IS 16714; maximum nominal size of aggregate 20 mm; workability 100 or 120 mm slump. Placing: chute / pump; supervision: fair / good; aggregate: angular / round. Maximum cement (OPC) content 450 kg/m^3; superplasticizer.
>
> Specific gravity: cement 3.15, fly ash 2.20, silica fume 2.20, GGBS 3.0, admixture 1.08, coarse aggregate (SSD) 2.74, fine aggregate (SSD) 2.65. Water absorption: CA 0.5%, FA 2%. Moisture content: nil. CA conforming to Table 7 and FA to Zone II (Table 9) of IS 383. Minimum cement content and maximum w/c as per Tables 3 and 5 of IS 456:2000.
>
> **Roll numbers starting with an odd digit**: chute, fair, angular. **Even digit**: pump, good, round.

Both cases are worked below side by side (IS 10262:2019).

Assumptions (justified):

- **Exposure: very severe** (coastal bridge deck exposed to sea-water spray and wetting/drying). IS 456 Table 5 (RCC): **max w/c 0.45, min cement 340 kg/m^3, min grade M35**. M50 satisfies the grade.
- **GGBS = 30% of total binder**: improves chloride resistance in marine exposure; within IS 455 limits.
- **Superplasticizer** dosage **1.0% of binder**, giving **20% water reduction**.

- Slump: **100 mm for chute** placing, **120 mm for pumping**.
- Entrapped air: **1%** for 20 mm aggregate.
- PPC specific gravity taken as the given cement value, 3.15.

Step 1: Target mean strength

f'_{ck} = larger of (f_{ck} + 1.65 s) and (f_{ck} + X), with s = 5 MPa and X = 6.5 MPa for M50. For **fair** supervision, s is increased by 1 MPa.

	Odd (chute, fair, angular)	Even (pump, good, round)
s (MPa)	5 + 1 = 6	5
f_{ck} + 1.65 s	50 + 9.9 = **59.9**	50 + 8.25 = **58.25**
f_{ck} + X	56.5	56.5
Target f'_{ck}	**59.9 MPa**	**58.25 MPa**

Step 2: Water/binder ratio

From Fig. 1 of IS 10262:2019 (53-grade curve): about **0.31** for 59.9 MPa and **0.32** for 58.25 MPa. Both are below the durability limit of 0.45, so these are adopted.

Step 3: Water content

Table 4: 186 kg/m^3 for 20 mm angular aggregate at 50 mm slump. For **rounded gravel reduce by 25 kg**. Add **3% per 25 mm** extra slump. Then the superplasticizer reduces it by 20%.

	Odd	Even
Base water	186	186 − 25 = 161
Slump correction	100 mm: +6% → 197.2	120 mm: +8.4% → 174.5
After 20% SP reduction	**157.7 kg/m^3**	**139.6 kg/m^3**

Step 4: Binder content

	Odd	Even
Total binder = water ÷ (w/b)	157.7 / 0.31 = **508.8 kg/m^3**	139.6 / 0.32 = **436.3 kg/m^3**
GGBS (30%)	**152.6**	**130.9**

	Odd	Even
PPC (70%)	**356.2** (< 450 ✓)	**305.4** (< 450 ✓)
Check min. 340 kg/m³	✓	✓
Superplasticizer (1%)	5.09	4.36

Step 5: Proportion of coarse aggregate

Table 5: volume of CA per unit volume of total aggregate = **0.62** for 20 mm aggregate with Zone II sand at w/c 0.50. Add **0.01 for every 0.05 decrease** in w/c. For **pumping, reduce by 10%**.

	Odd	Even
Correction	(0.50 − 0.31)/0.05 × 0.01 = +0.038	(0.50 − 0.32)/0.05 × 0.01 = +0.036
Volume of CA	**0.658**	0.656 × 0.9 = **0.590**
Volume of FA	0.342	0.410

Step 6: Mix calculations (per m³)

Volume (m³)	Odd	Even
a) Concrete	1.000	1.000
b) Entrapped air	0.010	0.010
c) PPC = mass ÷ (3.15 × 1000)	0.1131	0.0970
d) GGBS = mass ÷ (3.0 × 1000)	0.0509	0.0436
e) Water	0.1577	0.1396
f) SP = mass ÷ (1.08 × 1000)	0.0047	0.0040
g) All-in aggregate = a − (b + c + d + e + f)	**0.6636**	**0.7058**
h) CA = g × V_{CA} × 2.74 × 1000	**1196 kg**	**1142 kg**
i) FA = g × V_{FA} × 2.65 × 1000	**601 kg**	**766 kg**

Step 7: Correction for absorption (moisture content nil, so the aggregates are dry and will absorb water: CA 0.5%, FA 2%)

	Odd	Even
Dry CA = SSD ÷ 1.005	1190 kg	1136 kg
Dry FA = SSD ÷ 1.02	590 kg	751 kg
Extra water absorbed	17.7 kg	20.7 kg
Water to be added	**175.5 kg**	**160.3 kg**

Final trial mix per m^3 (site quantities, dry aggregates)

Material	Odd (chute, fair, angular)	Even (pump, good, round)
PPC	356 kg	305 kg
GGBS	153 kg	131 kg
Water (incl. absorption)	176 kg	160 kg
Fine aggregate (dry)	590 kg	751 kg
Coarse aggregate 20 mm (dry)	1190 kg	1136 kg
Superplasticizer	5.1 kg	4.4 kg
Free w/b ratio	0.31	0.32
Mix ratio (binder : FA : CA, SSD)	**1 : 1.18 : 2.35**	**1 : 1.76 : 2.62**

Durability notes for the coastal deck: w/b well below 0.45; GGBS reduces chloride diffusion; provide **nominal cover ≥ 50 mm** (very severe exposure, IS 456 Table 16); cure for at least 10–14 days because of the slag; confirm by trial mixes for slump, strength and chloride permeability.